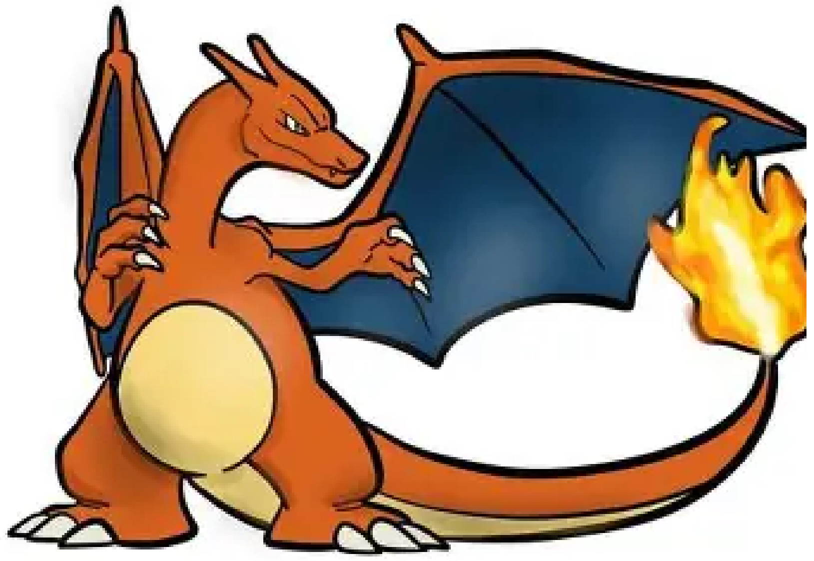
BATTLE SIMULATOR

Adam Yancey



WHAT THE GAME DOES

Core game loop

- Six different fighters, each with HP, stats, a type, and four moves.
- Turn-by-turn battles apply damage, accuracy, buffs, debuffs, healing, and status effects.
- The game decides a winner when one fighter reaches 0 HP.

Status effects	confusion, buffs, debuffs
Move accuracy	hit / miss
Outcome metrics	winner, turns, remaining HP
Battle choices	AI picks the best move

Measurements

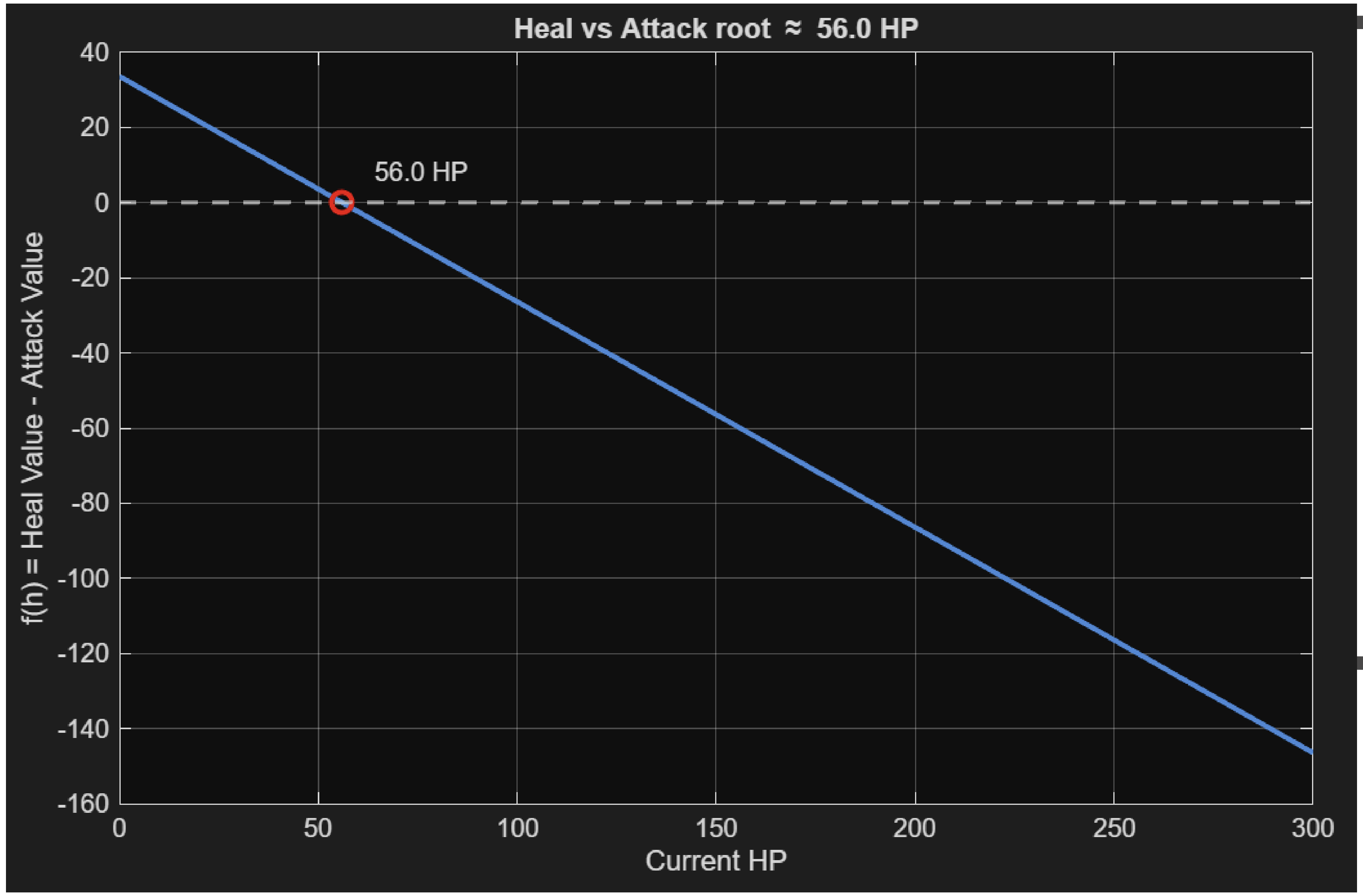
- damage calculation
- move success rate
- best move value by matchup
- win percentage by matchup
- average remaining HP at battle end

Root-Finding

How should AI know when to heal?

$$f(h) = EV_{\text{heal}}(h) - EV_{\text{attack}}(h)$$

- If $f(h) > 0$, healing gives more value than attacking.
- If $f(h) < 0$, attacking is better.
- A root of $f(h) = 0$ is the strategy threshold.



Example

$$EV_{\text{heal}}(h) = 0.15(300 - h)$$

$$EV_{\text{attack}} \approx 0.95\left(\frac{75}{70}\right)(36) = 36.6$$

$$f(h) = 0.15(300 - h) - 36.6$$

Bisection on [0,300] gives a threshold at 56 HP

Interpolation

$$\mathrm{EV}_{\text{heal}} = 0.15(300 - h)$$

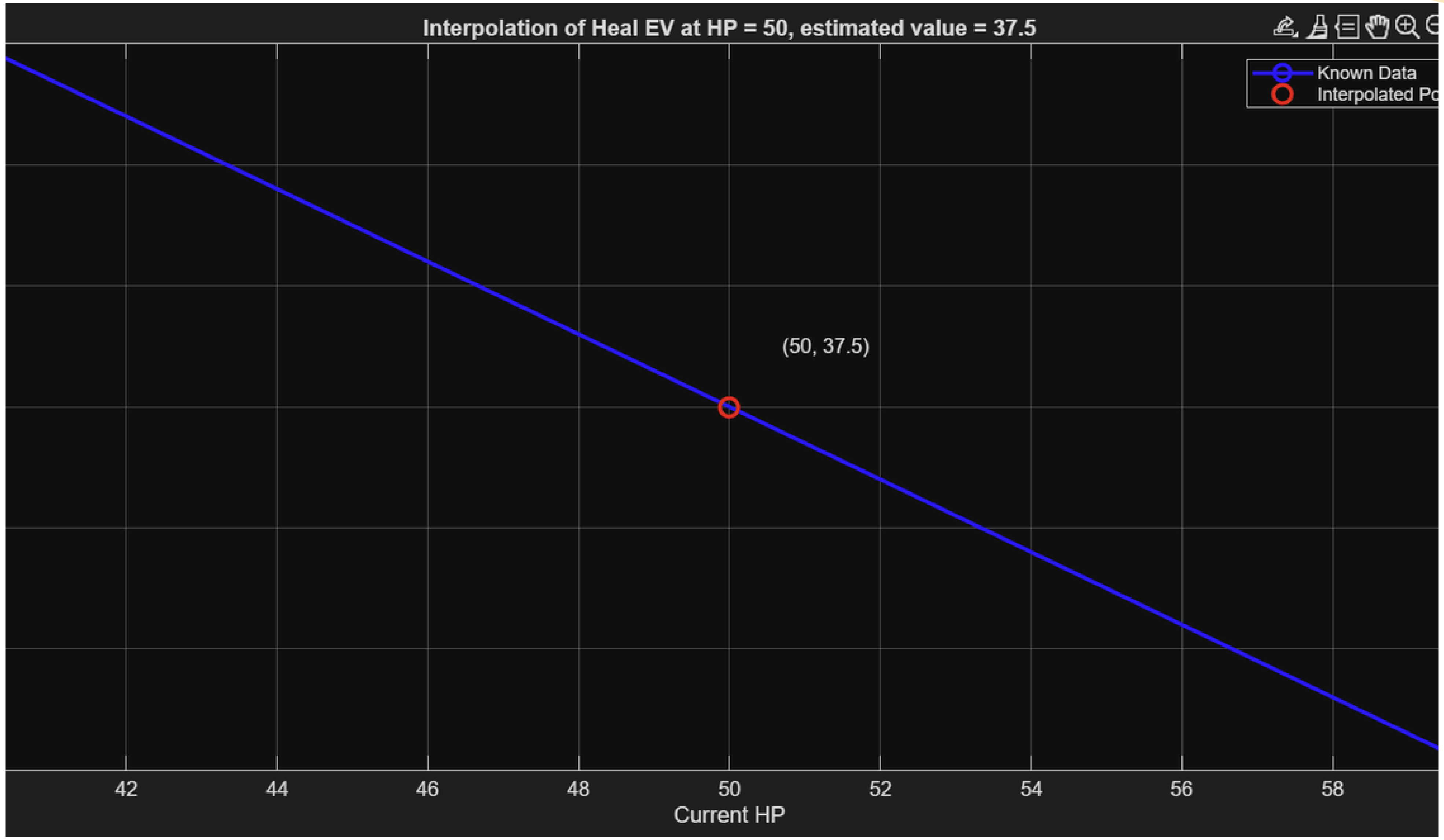
$$\mathrm{EV}_{\text{heal}} = 0.15(300 - 40) = 39$$

$$\mathrm{EV}_{\text{heal}} = 0.15(300 - 60) = 36$$

Interpolate at $h = 50$:

$$\mathrm{EV}_{\text{heal}}(50) \approx 39 + \frac{50-40}{60-40}(36 - 39)$$

$$\mathrm{EV}_{\text{heal}}(50) \approx 37.5$$



Interpolation is useful because it lets us estimate values between known data points without recomputing every possible HP value.

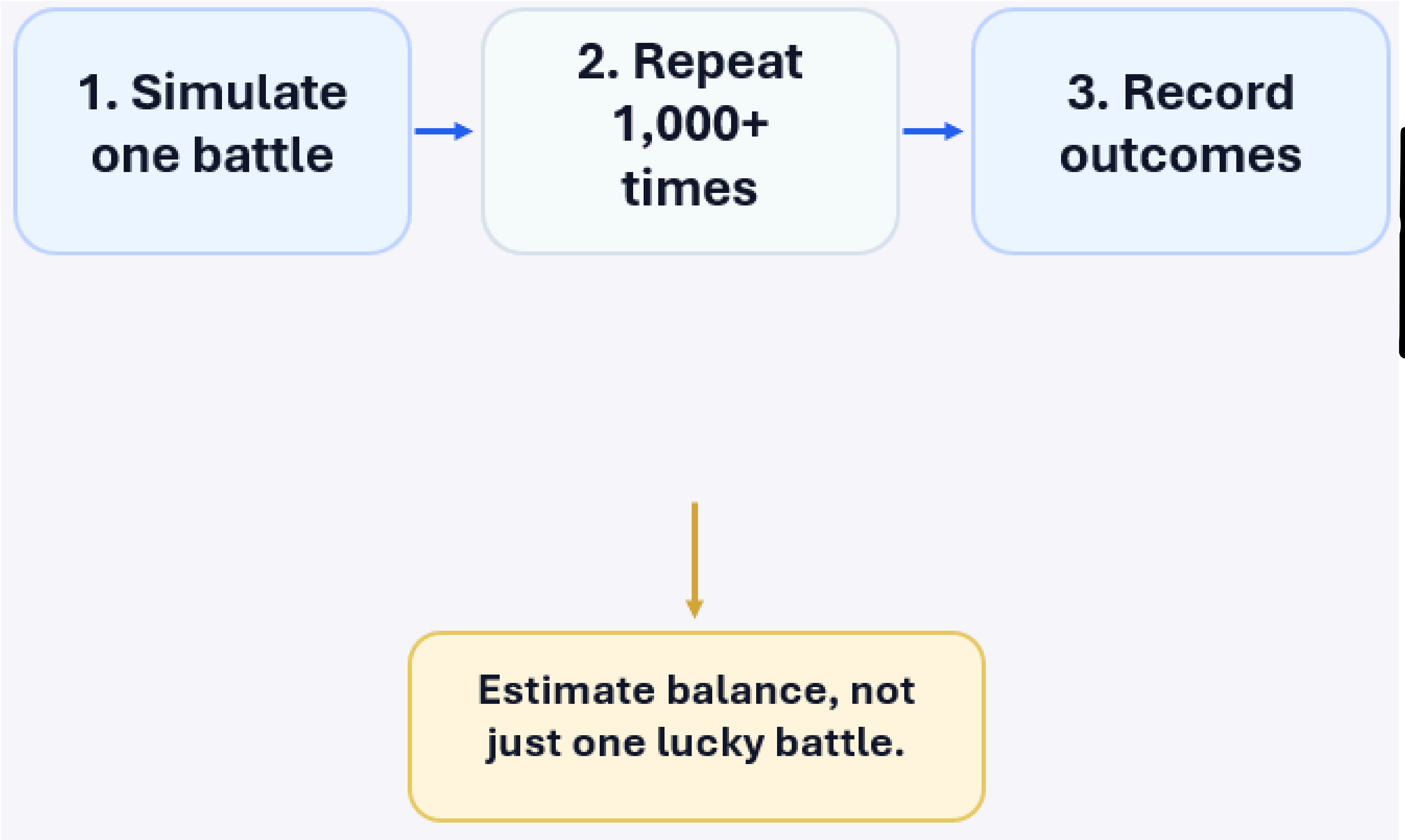
MONTE CARLO

What gets repeated?

- move accuracy rolls
- confusion self-hit chance
- multi-hit chains and healing rolls
- AI move choice or battle policy

What comes out?

- win rate by matchup
- average remaining HP
- average move value and turn count



THANK YOU